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Appendices

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Appendix A - Corpus Construction and Classifier

A.1 Pipeline Overview

#	Stage	Input size	Tools	Output
1	Download & segmentation	-	Python (BeautifulSoup)	~287,000 speeches with metadata
2	Dictionary filtering	~287,000 speeches	Custom dictionary (6 domains)	~32,500 speeches (11.3%)
3	Supervised ML classification	~32,500 speeches	Logistic regression (scikit-learn, threshold=0.65)	1,234 speeches (94% precision); 1,144 after removing 90 segmentation errors
4	Activism-category classification	1,144 speeches	Dictionary classifier with priority hierarchy (R)	1,056 speeches with topic (88 unassigned)
5	Window extraction	1,056 speeches	Sentence-level windowing (R)	Keyword windows for frame analysis
6	Keyness analysis	1,056 speeches	UDPipe italian-isdt-ud-2.5 + log2 TF ratio (R)	Distinctive lemmas per party-topic
7	STM (confirmatory)	1,056 speeches	stm R package (K per topic from searchK)	Latent cluster validation - see Appendix C

A.2 Corpus filtering dictionary

Category	Italian terms (lemma / variants)	English gloss
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Protest & dissent	protesta / proteste / protestare / protestano / protestando; dissenso; contestazione / contestazioni / contestare	protest; dissent; contestation
Collective action	mobilitazione / mobilitazioni / mobilitare; azione collettiva; partecipazione collettiva	mobilisation; collective action; collective participation
Movements & activism	attivismo; attivista / attivisti / attiviste; movimento / movimenti; movimento sociale / movimenti sociali	activism; activist/s; movement/s; social movement/s
Demonstrations	manifestazione / manifestazioni / manifestante / manifestanti; corteo / cortei; presidio / presidi; sit-in; sciopero / scioperi / scioperare; blocco stradale / blocchi stradali	demonstration/s; march/es; picket/s; sit-in; strike/s; road blockade/s
Claims & grievances	rivendicazione / rivendicazioni	demand/s; grievance/s
Boundary terms	rivolta / rivolte; sommossa / sommosse; insurrezione	revolt/s; riot/s; insurrection
Named groups & organisations	no tav / notav; no tap / notap; ultima generazione; extinction rebellion; fridays for future; centri sociali / centro sociale; casapound / casa pound; forza nuova; antagonisti / antagonista; black bloc / black block; antifa / antifascisti / antifascista; eco-vandali / ecovandali	No TAV; No TAP; Ultima Generazione; Extinction Rebellion; Fridays for Future; social centres; CasaPound; Forza Nuova; antagonists; Black Bloc; antifa / antifascists; eco-vandals

Note: Terms matched using case-insensitive exact string matching on tokenised speech text. All variants and inflexions listed are matched individually. Speeches must contain at least one term to be retained in the activism corpus.

A.3 Classifier Performance

The logistic regression classifier was trained on ~1,600 manually annotated speeches using 5-fold stratified cross-validation. Probability threshold set to 0.65 (precision-oriented).

Metric	Class 0	Class 1	Macro avg
Precision	0.72	0.78	0.75
Recall	0.70	0.71	0.71
F1-score	0.68	0.74	0.71

A.4 Annotation Guidelines Summary

Annotation criterion (the 'outside-in' rule): a speech is coded Class 1 if and only if debate centres on collective actors exerting political pressure from outside institutional channels.

Exclusion criteria:

- Internal parliamentary activity (procedural debate, committee reports, votes)
- Individual grievances or personal cases not linked to collective mobilisation (e.g. protesta as disagreement)
- Routine labour disputes (strikes over pay or contracts not framed as political protest or collective mobilisation)
- Polysemous keywords in non-activist contexts (e.g. occupazione as employment)
- Institutional actors engaging in activism without extra-institutional mobilisation
- Polysemous-term speeches (rivolta, movimento, blocco) were oversampled (~30% of training data) to improve robustness on difficult cases.

Appendix B - Activism-Category Classification

B.1 Classification Method

Each speech is assigned to one of 14 activism categories using a dictionary-based classifier with a priority hierarchy (01_topic_assignment.R). This approach is appropriate for classifying known protest referents, named movements, organisations, and activism types, rather than discovering unknown topics (Laver, Benoit and Garry 2003; Grimmer and Stewart 2013).

Classification logic:

- Keywords are defined as regex patterns applied to lowercased full speech texts
- Each topic is assigned a priority level (1 = most specific, 3 = broadest)
- When multiple topics match a speech, the topic with the lowest priority number wins
- Within the same priority level, the topic with the most keyword matches wins
- Speeches matching no keywords in any topic are coded as unassigned
- Priority 1 topics include named movements and organisations (Ultima Generazione, CasaPound, Hamas) which are inherently specific. Priority 3 topics (labour, students, police/security) are broader and would otherwise absorb speeches belonging to more specific categories.

B.2 Topic assignment dictionary (Stage 1)

Priority	Topic	Keywords / patterns (display form)
Priority 1 - Named movements and organisations (most specific)		
1	No TAV / No TAP	no tav, notav, val susa, valsusa, chiomonte, no tap, notap
1	Climate activists	ultima generazione, extinction rebellion, fridays for future, ecoattivisti, eco-vandali, imbrattato / imbrattare, blocco stradale / blocchi stradali
1	Pro-Palestine	palestina / palestinese / palestinesi, Gaza, Hamas, intifada, nakba, sionista / sionisti / sionismo
1	No Vax	no vax, novax, green pass, obbligo vaccinale, dittatura sanitaria
1	LGBTQ+	LGBT / LGBTQ / GLBT / GLBTQ, omofobia, transfobia, omotransfobia, DDL Zan, pride, unioni civili, omosessual*
1	Antisemitism	antisemit*, Shoah, Segre, ebrei / ebreo / ebraica / ebraismo, stella di Davide, sinagoga*, giornata della memoria, deportazione ebrei
Priority 2 - Clear movement categories		
2	Far right	casapound, forza nuova, neofascist*, estrema destra, saluto romano, squadrist* / squadrisimo, camerata
2	Antifascist	antifascist* / antifascismo, ANPI, partigian*, 25 aprile / venticinque aprile

2	Far left	centri sociali / centro sociale, black bloc, anarch*, antagonist*, autonomi, squat
2	Migrants & asylum	migranti, immigrat*, rifugiat*, profugh*, richiedenti asilo, sbarchi / sbarco, ONG, Sea Watch, Open Arms, Mediterranea, clandestin*, accoglienza, CPR, CIE, hotspot, respingiment*, decreto Salvini
2	Incarcerated persons	carcere / carceri, detenu*, penitenziario / penitenziari, sovraffollamento, 41-bis, ergastolo ostativo, celle, rivolta carceraria

Priority 3 - Broader categories (prone to overlap; flagged for review)

3	Students	studenti / studentesse, universitari, movimento studentesco, liceo / licei, occupazione studentesca, collettiv*
3	Labour	CGIL, CISL, UIL, sindacat*, scioper*, lavoratori / lavoratrici, operai / operaio, fabbric*, licenziament*, disoccupat*, precar*, vertenz*
3	Police / security forces	forze dell'ordine, polizia, carabinieri, agenti, divisa, questura / questure, polizia penitenziaria, pubblica sicurezza

Note: Keywords are listed in display form; the classifier applies case-insensitive regex patterns with word-boundary anchors where appropriate. Asterisks (*) indicate prefix truncation. Where multiple topics match, the lowest priority number wins; ties within the same priority level are broken by keyword frequency. Priority 3 categories (students, labour, police/security) are additionally flagged for manual review regardless of match strength owing to elevated false-positive risk.

B.3 Category Distribution in Final Corpus

Category	Priority	N speeches	% corpus	STM K
Incarcerated Persons	Priority 2	193	16.9%	K=2
Migrants	Priority 2	144	12.6%	K=3
Climate Activists	Priority 1	125	10.9%	K=2

Police / Security	Priority 3	113	9.9%	K=2
Far Right	Priority 2	87	7.6%	K=2
Labour	Priority 3	83	7.3%	K=2
Pro-Palestine	Priority 1	58	5.1%	K=3
Students	Priority 3	55	4.8%	K=2
No Vax	Priority 1	53	4.6%	K=2
Antisemitism	Priority 1	52	4.5%	K=3
Far Left	Priority 2	28	2.4%	n<30, excluded
Antifascist	Priority 2	26	2.3%	n<30, excluded
LGBTQ	Priority 1	22	1.9%	n<30, excluded
No TAV	Priority 1	17	1.5%	n<30, excluded
Unassigned	-	88	7.7%	-

Appendix C - searchK Results (STM K Selection)

C.1 Method

The number of latent topics K for each STM model was determined empirically using the `searchK()` function in the `stm` R package (Roberts et al. 2014), evaluated across K = 2 to 7. K was selected based on held-out likelihood, the primary criterion, which measures how well the model predicts unseen words. Semantic coherence was used as a secondary criterion. Topics with fewer than 30 speeches were excluded from STM analysis.

C.2 Recommended K per Topic

Topic	Selected K	N	Interpretation
		speeches	

Incarcerated Persons	K=2	193	Reflects binary divide between humanitarian framing by AVS/PD and punitive security/delegitimisation framing by FDI/LEGA
Climate Activists	K=2	125	Reflects opposition between security/delegitimisation framing by FDI/LEGA and legality framing by AVS/PD/M5S
Police / Security	K=2	113	Reflects opposition between accountability-based legality frame by AVS/PD and order/protection security framing by FDI/LEGA/FI
Labour	K=2	83	Reflects division between solidarity/legality framing by AVS/PD and delegitimisation/security framing by FDI
Far Right	K=2	87	Reflects primary opposition between legality/delegitimisation framing by centre-left parties and limited engagement from right-wing parties, with M5S introducing a secondary security dimension
Students	K=2	55	Reflects opposition between security/delegitimisation framing by FDI and legality framing by PD.
No Vax	K=2	53	Reflects polarisation between delegitimisation framing and a security/legality cluster, though discourse appears more fragmented than other K=2 topics.
Migrants	K=3	144	Reflects three-way split between securitarian framing by FDI, national identity/security

			framing by LEGA, and humanitarian/legality framing by AVS/PD
Pro-Palestine	K=3	58	Reflects three dimensions: humanitarian/legality framing by AVS/PD, security/delegitimation by FDI, and legality framing by M5S
Antisemitism	K=3	52	Reflects distinction between morality framing by FDI, security by FI, and delegitimation framing by AVS and PD

Note: Most activism debates show binary legitimacy conflicts (K=2). Topics linked to identity, geopolitics, or migration show three-dimensional discourse structures (K=3).

Appendix D - STM Results (Confirmatory)

D.1 Model Specification

STM was run separately for each topic with $n \geq 30$ speeches, using the topic-specific K values from searchK (Appendix C) and party affiliation as a prevalence covariate. STM was applied to keyword windows rather than full speech texts. The purpose is confirmatory: to verify that meaningful latent thematic clusters exist within each topic's speeches, and to triangulate the framing patterns identified through keyness analysis. STM results are not reported as primary findings.

Model parameters:

```
prevalence = ~party_clean; init.type = 'Spectral'; max.em.its = 75
```

Models were estimated using the stm R package

Appendix E - Keyness Analysis: Original Italian Lemmas

E.1 Method

Keyness analysis identifies vocabulary disproportionately associated with each party's discourse relative to all other parties discussing the same activism category. The distinctiveness score is computed as:

$$\text{distinctiveness} = \log_2((\text{tf_party} + 0.0001) / (\text{mean_tf_others} + 0.0001))$$

Lemmatisation: UDPipe italian-isdt-ud-2.5-191206. Content words only (NOUN, VERB, ADJ, ADV). Min frequency: $n \geq 2$. Min speeches per party-topic cell: $n \geq 5$. +EUROPA and AZIONE excluded (insufficient n per cell). MISTO excluded (heterogeneous group).

The figure in the main thesis body presents English translations for accessibility. The original Italian lemmas are documented below.

E.2 Top Distinctive Lemmas per Party per Topic (Italian)

See `keyness_top15_wide.csv` in the GitHub repository for the full output. The table below reproduces the top 15 lemmas per party-topic cell as used in the analysis.

Top Keyness Lemmas by Party and Activism Topic (Italian Terms)

Most overrepresented lemmas in each party-topic corpus relative to other parties discussing the same topic

topic	AVS	PD	MSS	IV	FI	LEGA	FDI	LEU
antisemitism	mese, alleato, contare, euro, inutile, mezzo, tenda, chiudere, storia, comunque, attacco, milione, invece, consiglio, via	riguardare, interno, malessere, vivere, sportivo, elettorale, impedire, militante, protestare, serio, sicuramente, simbologia, studente, appunto, disagio			divisa, esistere, ostruzionismo, terrorista, antitesi, aperto, boicottaggio, conoscere, funesta, gennaio, inquietante, negazionista, novecento, polizia, presunto		sinistra, ebraico, inneggiare, israeliano, aprile, caratterizzare, continuare, partecipare, scendere, senatore, susseguire, accademico, articolo, censura, centrodestra	
climate_activists	sovraffollamento, fascista, elemento, impresa, record, accordo, allevatore, applauso, asilo, atteggiamento, cancellare, casso, climafregghista, comitato, dispositivo	progressista, democratica, aspettare, appello, circostanza, disperazione, spingere, accanimento, adiacenza, cire, contadino, contrario, esattamente, figura, rientrare	emarginazione, investire, purtroppo, arresto, crimine, fallimento, giornalista, lanciare, bavaglio, costante, garantismo, scusa, utile, volgere, addestrare			pastore, precludere, costare, soffiare, telecamera, commento, limite, dibattito, eco, azione, valere, ferrovia, probabilmente, periodo, racket	dichiarare, proprietà, teatro, artistico, attendere, candidare, conforme, diametralmente, eleggere, influire, nervo, occupato, occupazioni, opposto, paladina	
far_right	credit, emendamento, everisivo, fondante, tax, anzitutto, bocciare, buttare, climatico, corpo, dibattito, ennesimo, identificare, indecente, persino	sede, quartiere, attività, grave, interpellanza, appartenere, episodio, svolgere, affitto, clima, peraltro, squadrista, trattare, vero, aspetto	gestire, affermare, amministratore, condividere, decennio, incinto, inerme, mattino, operazione, riuscire, terrificante, trasparenza, ulteriore, vetrina, occupazione	funzionare, finora, negozio, tamponare, scendere, sera, fondamentale, soggetto, pensiero, magari, capire, aspettare, atto, violento, libertà				
labor	sindacale, strumento, climatico, organizzazione, padrone, produrre, riguardare, spesso, studente, autoritario, come, dopo, elemento, garantista, licenziamento	pensiero, conquistare, disabile, pericolo, serio, libertà, picchiare, rischio, sfruttamento, aspettare, lottare, praticamente, contributo, media, forma	bomba, esercitare, partecipare, regalare, sottosegretario, agosto, cittadinanza, coprire, derivare, diffuso, dosso, famoso, ideale, macchina, mancato				record, lingua, mai, associazione, annuncio, categoria, confronto, divertare, migliaio, perdita, quartiere, ramo, tavolo, verità, xviii	
migrants	meno, ricorso, software, ammettere, climatico, danno, denuncia, detentivo, deterrente, garantismo, giornalista, impossibile, indelebile, membro, ritirare	giudice, messaggio, ragione, corso, video, asilo, campagna, estremo, partito, semplice, collettivo, comportamento, consenso, crinale, giustizia	permesso, soggiorno, accusa, aiutare, concreto, conduttore, datore, duro, espressione, inevitabilmente, insegnare, liberticido, pugno, silenziare, temporaneo			sinistra, sindaco, vostro, autorizzazione, avviare, bade, esterno, fascista, favorire, frontiera, libico, ministero, normale, opporre, porta	calpestare, fregare, giurato, tentare, volte, fallire, quarantena, realizzare, accedere, adeguatamente, autonomamente, bando, bonus, carcerario, caserma	esperienza, locale, preparare, intervenire, comunità, articolo, esame, riunione, esistere, bastare, legge, motivo, criminalizzare, costituzione, penale
no_vax	dentro, fornire, gratuito, nemico, occasione, politicaipolitico, punire, restare, evento, giovane, intervenire, cercare, indietro, raduno, musicale	pensiero, bavaglio, ecco, insomma, pericoloso, provvedimento, città, idea, preventivo, reale, studente, abitazione, cronaca, dibattito, evidentemente	violento, condotta, formazione, leader, osservare, sfociare, alto, assalto, avviso, distinguere, efficace, esempio, estremismo, giudiziario, giustizia	comunità, detenuto, eversivo, polizia, comprensibile, condividere, destare, elemento, grazie, matrice, ministero, numeroso, personale, rigoroso, succedere				
police_security	militare, circa, fenomeno, gestire, minaccia, ponte, spontaneo, aumentare, clima, commissariato, determinare, dirigere, domani, esercitazione	disagio, proiettile, sproporzione, azienda, condividere, crisi, cultura, ecco, finire, forse, impegno, incontrare, informativa, ragionare, riferimento	soldato, arrivo, foglio, risorsa, spiegare, accordo, adottare, aggiuntivo, azzerare, costituzionalmente, imprimere, motivo, naturalmente, politicaipolitico, prefetto		sopportare, autorizzato, dispiacere, ottenere, percezione, abnegazione, carabinieri dico, massimo, onore, opposizione, procedimento, stenografie, vergognoso, esistere	aprire, comprensione, concertare, disperazione, futuro, pace, riproporre, ristorante, manifestazioni, garantire, intero, cercare, solidarietà, scontro, agente	armato, amore, silenzio, sindacale, animare, compito, debole, eletta, immagine, indifese, insignire, madrepatría, osservare, passione, patrio	
incarcerated_persons	climatico, cambiamento, desistenza, identificare, indicare, abbraccio, corte, definitivo, depurato, impiego, incentivare, incesante, inciso, inconcepibile, indignazione	urbano, resistere, risultato, catena, contraddittorio, ente, innocente, nemico, odioso, segreto, allarme, amico, ampio, attivo, civico	accedere, bocca, bugia, corrotto, formulazione, incidere, peso, popolo, riunione, soffocare, tecnico, abominevole, adempiere, antimafia, applicabile	prezzo, sproporzione, trasmettere, vicinanza, accusa, arte, custodia, equilibrio, ignobile, maiale, mozione, operativo, porcellaia, procura, ricercatore	convocare, abusivamente, confermare, giornalismo, immagine, ordinario, riguardo, pomeriggio, rappresentare, immobile, valore, forza, opposizione, mandare, condividere	body, cam, euro, certamente, investimento, superare, disporre, legale, tutela, guardia, agente, oltre, garanzia, assistere, milione	mortificare, flagellare, protocollo, supremo, eterodiretto, furto, certezza, circostanza, comunicazione, disordine, domare, domenica, durato, equipaggiamento, fermezza	
pro_palestine	violazione, appunto, cartello, chiudere, davvero, mobilitazione, accesso, agire, agricoltore, ambientalista, artista, buonista, cattivista, contatto, disegualianza	accettabile, aiuto, nemmeno, sciopere, aprire, coprire, criminalizzazione, esercito, grido, legalizzare, legare, nave, sentimento, servire, crimine	stampa, esame, spazio, assolutamente, decidere, significare, ambiente, ascoltare, culturale, dialogo, modello, prendetevelare, punire, soldato, veramente				agente, anti, argomento, tentare, affermare, ambiguità, avanti, benzina, bloc, brigare, coro, fase, film, fuoco, immagine	
students	destra, classe, dedicare, lista, nemmeno, organizzazione, appello, inerme, investire, liceo, masco, pagare, piano, povero, riflessione	comune, controllo, aprire, clima, mozione, residenza, zona, accesso, adottare, autonomia, autorità, circolazione, conseguenza, fermare, mobilità	chiaro, tagliare, anziché, combattere, curioso, insegnare, storia, alquanto, attenzione, chiaramente, costituzione, dialogare, dissidente, ecetetera, fascista				burocratico, competenza, conferenza, disagio, rivolta, rivoltoso, possibile, lavorare, percorso, dimostrare, incredibile, neanche, bastare, come, purtroppo	

Cells report the highest-scoring lemmas by log-ratio keyness within each party-topic combination, comparing each party's vocabulary with that of all other parties discussing the same topic. Terms are shown in original Italian. Blank cells indicate insufficient observations or no stable results after minimum frequency thresholds were applied.

Table: Camilla Gazzola • Created with Datawrapper